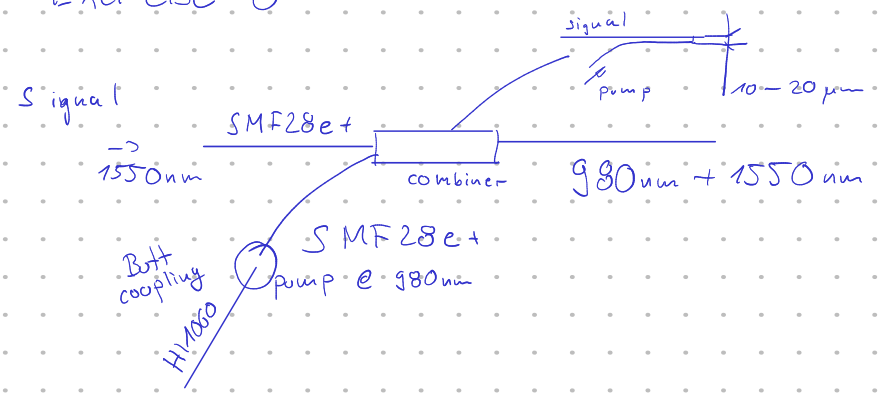
# AOFC Exercise 8

09.12.2025




a) HI 1060 $d_{1060} = 3{,}4\,\mu m$ $A_N^{1060} = 0.21$

SMF 28e+ $d_{28} = 8{,}2\,\mu m$ $A_N^{28} = 0.14$

$$V_{1060} = 4 \cdot a\, A_N = \frac{2\pi}{0.98\,\mu m} \cdot \frac{3{,}4\,\mu m}{2} \cdot 0.21 = 2.29$$

$$\frac{\omega}{a} \approx 0.65 + 1.619\, V^{-3/2} + 2.879\, V^{-6} \qquad \left(0.65 + 1.619 \cdot 2.29^{-3/2} + 2.879 \cdot 2.29^{-6}\right) \frac{3{,}4\,\mu m}{2}$$

$$\omega_{1060} = 2\,\mu m$$

$$V_{28} = \frac{2\pi}{0.98\,\mu m} \cdot \frac{8.2\,\mu m}{2} \cdot 0.14 = 3.68$$

same for every wavelength! (assumption!)

$$\omega_{28} \approx \left(0.65 + 1.619 \cdot 3.68^{-3/2} + 2.879 \cdot 3.68^{-6}\right) \cdot \frac{8{,}2\,\mu m}{2} = 3{,}61\,\mu m$$

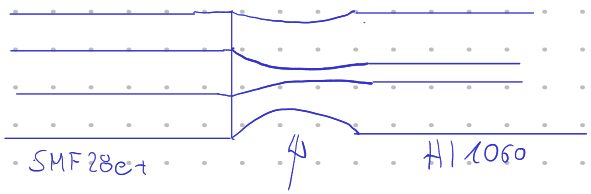
$$\eta = \left(2 \cdot \frac{\omega_1 \cdot \omega_2}{\omega_1^2 + \omega_2^2}\right)^2 \exp\left(-2 \cdot \frac{\Delta x^2}{\omega_1^2 + \omega_2^2}\right)$$

$$\eta = \left(2 \cdot \frac{3{,}61\,\mu m \cdot 2\,\mu m}{3{,}61^2\,\mu m^2 + 2^2\,\mu m^2}\right)^2 \exp\left(-2 \cdot \frac{0.5^2\,\mu m^2}{3{,}61^2\,\mu m^2 + 2^2\,\mu m^2}\right)$$

$\eta = 0.698$ very bad coupling!

$\eta_{log} = 10 \cdot \log_{10}(0.698)$

$= -1.561\ dB$




Taper for reducing loss!

b) from Agrawal:

$P = 1 - A \qquad I \propto |E|^2$

$$\Gamma^2 = \frac{P_{core}}{P_{total}} = \frac{\int_0^a |E_x|^2 \rho \, d\rho}{\int_0^\infty |E_x|^2 \rho \, d\rho} = 1 - \exp\left(-\frac{a^2}{\omega^2}\right)$$

$\not\propto \quad E_x = e^{-\frac{\rho^2}{\omega^2}} e^{i\beta z}$

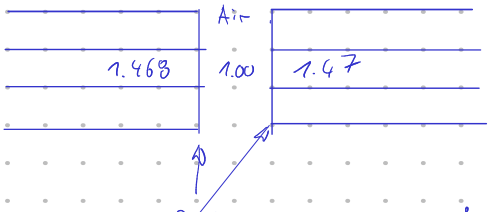
$$\Gamma^2 = 1 - \exp\left(-\frac{2a^2}{\omega^2}\right) = 1 - \exp\left(\frac{2 \cdot (4.1\,\mu m)^2}{3.485\,\mu m^2}\right)$$

$\Gamma^2 = 0.93$

93 % of the field is inside the core! ($LP_{01}$)

Higher will interact stronger with the cladding!

c)



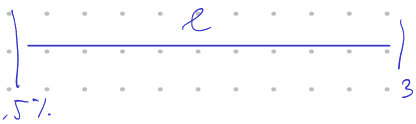

Reflexions! field hits the interface @ 90° = Θ

$$R_1 = R_s = R_p = \left| \frac{n_1 \cdot \cos\theta_2 - n_2 \cdot \cos\theta_1}{n_1 \cdot \cos\theta_1 + n_2 \cdot \cos\theta_2} \right|^2 = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2 = \left| \frac{1.468 - 1}{1.468 + 1} \right|^2 = 0.035$$

$T_1 = 1 - R_1 = 0.96$

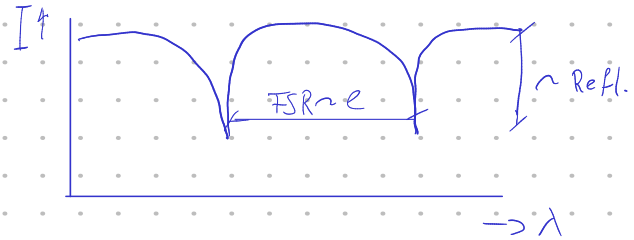
$$T_{tot} = T_1 - R_2 = T_1 - \left| \frac{1 - 1.47}{1 + 1.47} \right|^2 = 0.928$$

$R_{total} = 0.072$




-> Be aware of reflexions at fiber interfaces!

High losses at fiber coupling can build a fabry-perot-interferometer



d) $V = \frac{2\pi}{0.98\mu m} \cdot 4.1\mu m \cdot 0.14 = 3.68$

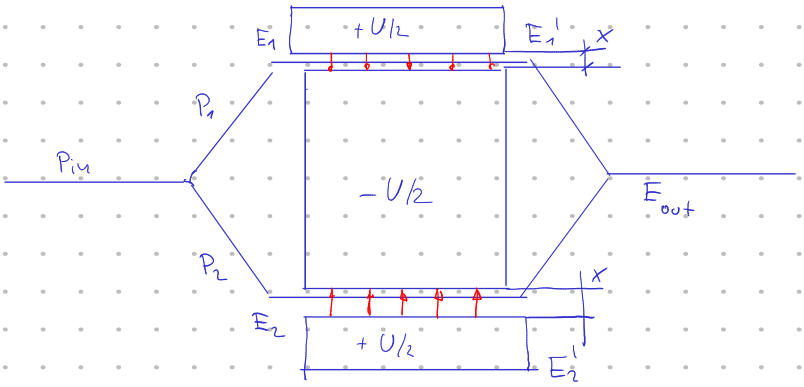
→ From Diagramm: $LP_{01}$, $LP'_{11a}$, $LP_{11b}$

→ MFD of $LP_{11a,b}$ is higher, and is degrading amplifier performance

→ for reducing power inside higher order modes: wind up fiber around a small diameter.

Exercise 14

a) Transmission Ratio as a function of $p = \frac{P_1}{P_2}$ & $\Delta\gamma, -\Delta\gamma$




1) Power Splitter

$p = \frac{P_1}{P_2}$ $\quad P_{IN} = P_1 + P_2$

$P_{IN} = p \cdot P_2 + P_2 \quad / : P_2$

$\frac{P_{in}}{P_2} = p + 1$

$\frac{P_2}{P_{IN}} = \frac{1}{p+1}$ $\quad \frac{P_1}{P_{IN}} = \frac{p}{p+1}$

Phase modulation: how is the E-Field connected to the Power?

$P \sim I$ $\quad I \sim |E|^2$

$P_{1,2} = (E_{1,2})^2$ $\quad E_{1,2} = \sqrt{P_{1,2}}$

$$E_1 = \sqrt{\frac{P}{P+1}} \cdot E_{IN} \qquad E_2 = \sqrt{\frac{1}{P+1}} \cdot E_{IN}$$

2) Phase Modulators

Upper Branch: $\Delta\varphi$ phase shift

lower Branch $-\Delta\varphi$ phase shift

$$E_1' = E_1 \cdot e^{i\Delta\varphi} = E_{IN} \cdot \sqrt{\frac{P}{P+1}} \cdot e^{i\Delta\varphi}$$

$$E_2' = E_2 \cdot e^{-i\Delta\varphi} = E_{IN} \cdot \sqrt{\frac{1}{1+P}} \cdot e^{-i\Delta\varphi}$$

3) Power Combiner

$$\frac{E_1}{\sqrt{2}} + \frac{E_2}{\sqrt{2}} = E_{out} \rightarrow \frac{1}{\sqrt{2}} \text{ due to Energy conservation}$$